MODULE 6, LESSON 3

AI 101 FOR MECHANICAL ENGINEERS

Baselines prevent empty success claims

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Outline

1. Baseline Methodology
2. Error Interpretation
3. System Reproducibility

Section 1
Baseline Methodology

Purpose: Establish simple comparison methods evaluated under identical conditions.

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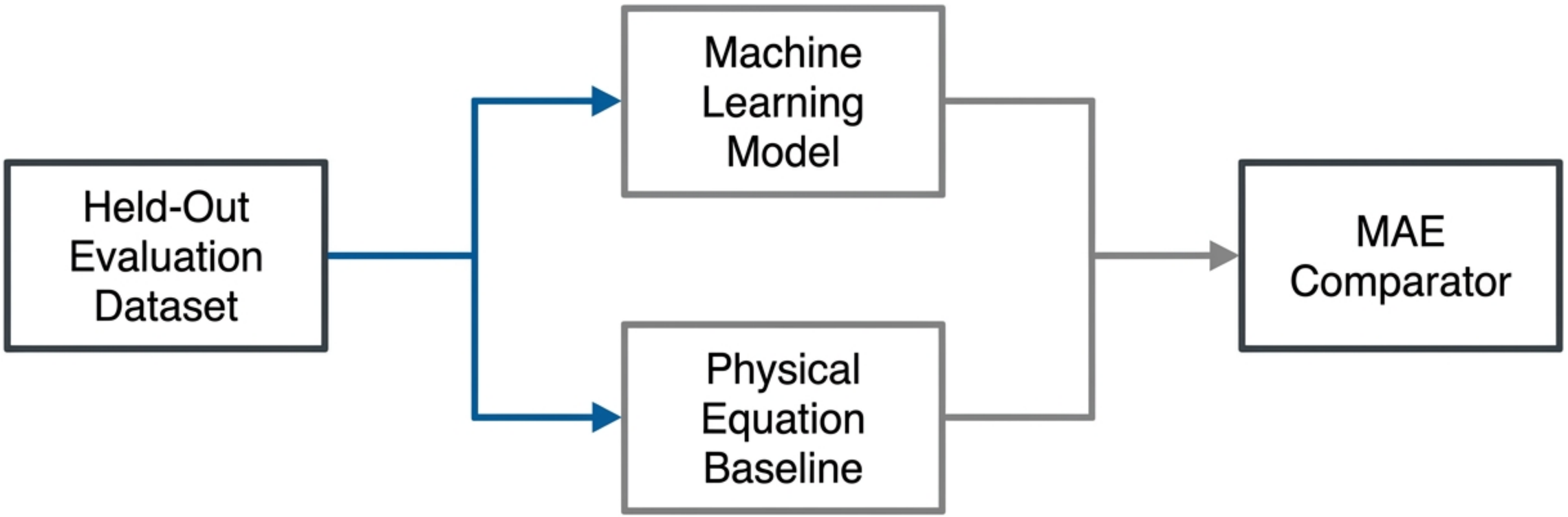
The Baseline Evaluation Matrix

- A baseline is a simple comparison approach.
- It must use identical evaluation examples.
- It must use identical information availability.

Regression Tasks	Classification Tasks
Predict the training mean OR Use a trusted physical relationship.	Compare with a majority-class rule AND Apply appropriate fault-sensitive metrics.

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Fair Comparison Architecture



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Section 2
Error Interpretation

Purpose: Analyze model error metrics against established physical baselines.

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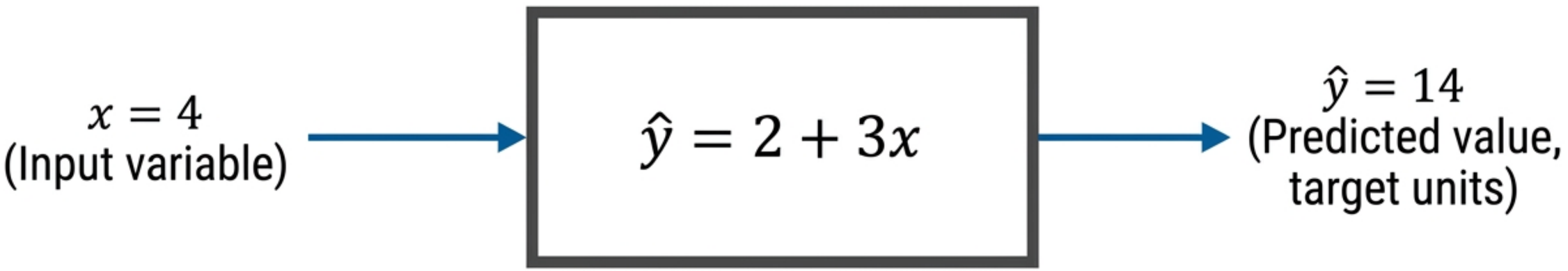
Model vs. Physical Baseline Diagnostic

- Evaluated on identical held-out test data.
- Cost and operating-range evidence remain critical factors.
- Uses AI is not a valid performance claim.

System Type	Machine Learning Model	Physical Baseline
Error Magnitude	MAE = 1.2 kW	MAE = 0.8 kW
Evaluation Outcome	Higher Error (+ 0.4 kW)	Lower Error (Superior)

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Inference and Parameter Fixing



Inference operations keep fitted coefficients fixed.
Learned parameters: Intercept (2) and Slope (3).
Model demonstrated no MAE benefit.
Evaluated operating range limits the system claim.

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Required Interpretative Context

An isolated error **metric requires documented context**:

- [x] 1. Target unit specification.
- [x] 2. Evaluated population definition.
- [x] 3. Data split methodology.
- [x] 4. Measurement uncertainty limits.
- [x] 5. Baseline error for fair comparison.

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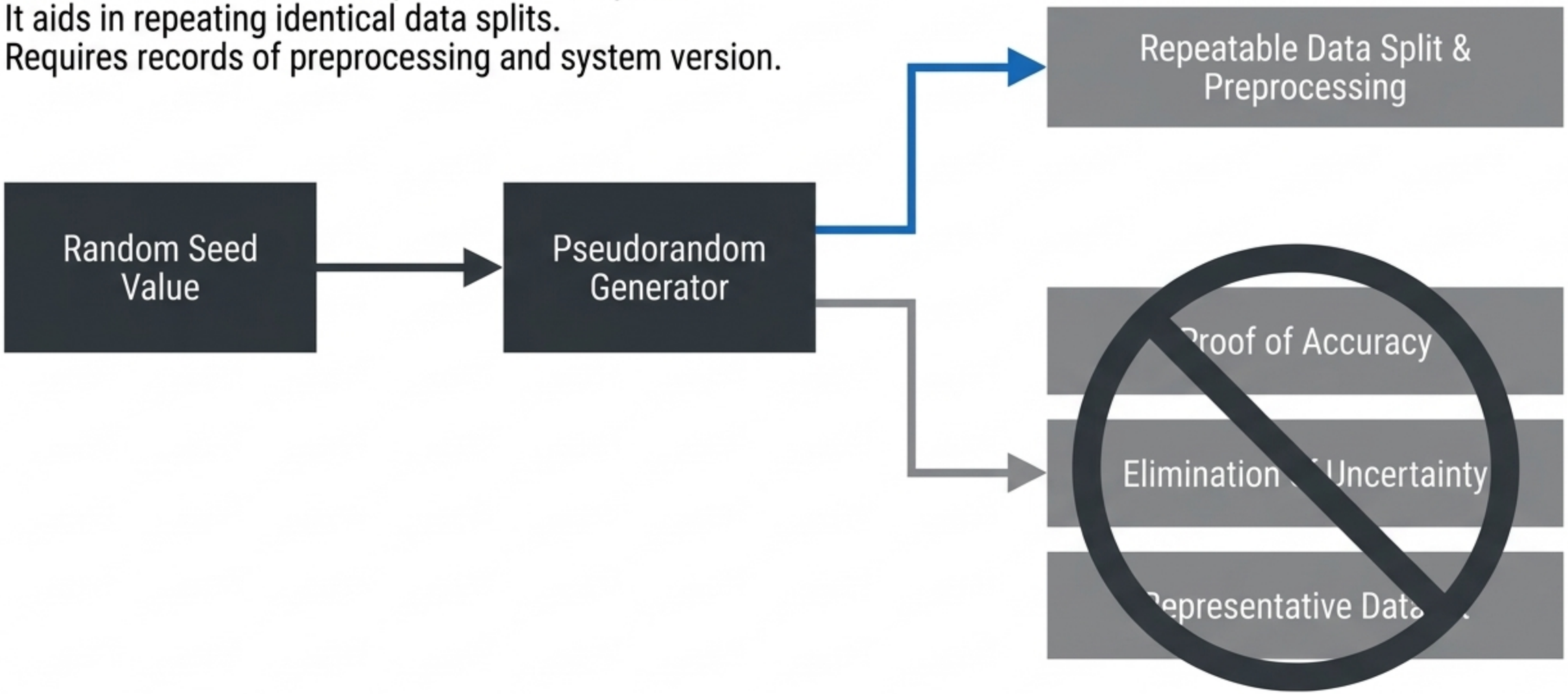
Section 3
System Reproducibility

Purpose: Define the technical requirements and functional limits of systemic validation.

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Random Seed Capabilities

A random seed initializes a pseudorandom procedure.
It aids in repeating identical data splits.
Requires records of preprocessing and system version.



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The Evidentiary Reference Matrix

System claims require specific, verifiable evidentiary parameters.

Evidentiary Parameter	Established Engineering Proof
Same-set baseline	A fair performance comparison.
Target units	The physical meaning of error magnitude.
Split IDs	Traces which examples informed each stage.
Random seed	Ensures a repeatable random procedure.

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Procedural Integrity Requirements

Comparison Prerequisites:
Require common evaluation examples, identical inputs, and stated metrics.

Training Limits:
A training score is invalid for held-out comparisons.

Metric Hierarchy:
Lower error magnitudes are strictly superior given matched units.

Reproducibility Limits:
Repeatability under stated conditions does not equal procedural validity.

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Key Points

1. ML models must be evaluated against standard physical baselines using identical data.

2. Error metrics require comprehensive contextual parameters to prove technical benefit.

3. Random seeds ensure procedural repeatability, not physical accuracy or validity.

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